

SDC Assignment 4 - Implement Kalman Filter

Ying Pei Lin

Spring 2026

Code Explanation

Filtered Path Result

Design of Observation Matrix (C)

Design of Covariance Matrices (Q, R)

Analysis: $Q=0.001$, $R=1000$

Analysis: $Q=1000$, $R=0.001$

Impact of Q and R on Filter Output